

# ETHAN (TAEHOON) LEE

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## EDUCATION

### Carnegie Mellon University

Master of Software Engineering (4.0/4.0 GPA)

Pittsburgh, PA

Expected Dec 2027

### Emory University

BS in Computer Science (4.0/4.0 GPA, Summa Cum Laude, 5x Dean's List)

Atlanta, GA

May 2026

## WORK EXPERIENCE

### Founding Software Engineer

Huddle Surety (Techstars Pre-Accelerator '26)

Atlanta, GA

Sep 2025 – Present

- Launched an AI platform replacing manual surety processing at 10x speed, running 100+ bonds daily across 5 agencies
- Developed the core Go REST API spanning 200+ endpoints with Redis caching and Temporal workflow orchestration
- Engineered multi-stage RAG pipelines with cross-document context retrieval generating bond forms at 95% accuracy
- Shipped a role-based frontend delivering distinct surety agent and contractor UX using Next.js, TypeScript, and tRPC
- Architected end-to-end context propagation using OpenTelemetry, enabling per-client usage metrics and cost analysis

### Software Engineering Intern (Cloud Infrastructure)

UKG

Atlanta, GA

May 2025 – Aug 2025

- Automated infrastructure operations in a hybrid cloud environment spanning 4 on-prem datacenters and 3 GCP regions
- Built a TLS lifecycle monitor in Go, packaged as a Helm chart and scheduled as a Kubernetes CronJob, replacing manual healthchecks of 200+ OpenStack VMs during annual certificate renewal and reducing renewal time by 50%
- Partnered with Grafana Labs engineers to create database synthetic monitoring using Grafana K6 and Blackbox Exporters
- Developed a unified Go REST API to scatter-gather network metrics across 5 Kubernetes clusters using OpenTelemetry

### Software Engineering Intern (Full Stack)

eStreamly

Atlanta, GA

Jun 2024 – Aug 2024

- Engineered frontend performance and deployment tooling for a shoppable livestream platform serving 100+ brands
- Built client-side caching and a reactive global state store in video engagement UI, cutting redundant API calls by 40%
- Created a multi-stage container build and release pipeline on GCP Cloud Build, improving deployment speed by 35%

## PROJECTS

### Kubernetes Homelab & Contribution | Kubernetes, Linux, FluxCD, Cilium, Grafana, OpenTelemetry

headlamp.thdxg.dev

- Designed a two-node Kubernetes cluster on Raspberry Pi serving 20+ projects including personal website and dashboards
- Automated CI/CD using FluxCD and provisioned public ingress using Cilium CNI, Gateway API, and dynamic DNS
- Authored the Korean Kubernetes documentation on pod scheduling and eviction as part of SIG Docs localization efforts

### DooleyOnline: Emory Student Marketplace | Go, Next.js, React, TypeScript, PostgreSQL, AWS

github.com/dooleyonline

- Launched an online marketplace for Emory students to exchange secondhand items, reaching 150 users on campus
- Implemented real-time buyer-seller chat over WebSockets with persisted message history and S3-backed image uploads

### Macterm: Terminal Emulator with Persistent Multiplexing | Swift, SwiftUI, C

github.com/thdxg/macterm

- Built a native macOS terminal with SwiftUI and libghostty, gaining 6K+ downloads and 20 open-source contributors
- Implemented session persistence, declarative layout configuration, fuzzy command palette, and configurable hotkeys

### Spotify Warped: HCI Research & Publication | Python, JavaScript, React

dl.acm.org/doi/10.1145/3772318.3790875

- Published to ACM CHI 2026 on how Spotify's algorithmic feedback loops shape listening behavior and data engagement
- Created high-throughput data processing pipelines in Python to ingest 100 GB of Spotify streaming data from 60 users

## SKILLS

- **Languages:** Go, TypeScript, JavaScript, Java, C, Rust, Python, SQL, Swift, x86 Assembly
- **Technologies:** Kubernetes, Linux, Docker, AWS, GCP, PostgreSQL, Redis, OpenTelemetry, Temporal, Prometheus, Grafana